Leveraging Technology to Combat Food and Packaging Waste on Campus: A Sustainable Approach

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Leveraging Technology to Combat Food and Packaging Waste on Campus: A Sustainable Approach

Selected Issue: Food and Packaging Waste on Campus

Waste from food and packaging has become a serious problem on campus, yet many students don't realize how big the issue has gotten. If you've ever walked past the café after class, you've probably seen stacks of empty bottles, leftover food, and discarded paper plates. This waste is a reflection of how seriously (or not) we take sustainability, and it's more than just an inconvenience. It's not just about overflowing trash cans and dirty tables; it points to a bigger issue with our responsibility to protect the environment. We're not only creating this waste, but we're also paying for its removal through our tuition. And despite all the talk about caring for the environment, we continue to throw away half of our meals without a second thought. This disconnect between what we say and what we do makes it hard to take our dedication to sustainability seriously.

Those who spend a lot of time on campus—such as tenants of student housing, students who attend classes back-to-back, and anyone who depends on the cafeteria for meals—are the most impacted. Disposable containers are frequently given out by cafeterias and food vendors, but because student life moves quickly, many people forget to properly dispose of them. Cleaning crews are overworked as a result, and college "Go Green" teams, who frequently plan awareness campaigns, don't see any long-term results because students tend to disregard these efforts quickly. It is obvious that conventional approaches, such as posters and awareness gatherings, are insufficient to address the issue.

The issue is rooted in behavior rather than a lack of awareness. Although they frequently put convenience ahead of sustainability, students are not naturally irresponsible. Although it should be, properly disposing of food waste is rarely the first thing that comes to mind in the rush between courses. Therefore, the task is to make sustainable activities simple, interesting, and fulfilling.

This is where technology could play a transformative role. Consider a smartphone app that keeps tabs on students' environmentally conscious decisions, awards them with points, and might even let them compete on a leaderboard. Making the transition to sustainability could be as fulfilling as getting alerts to switch off lights. Through the integration of technology, we can motivate students to embrace more sustainable practices that align with their everyday routines.

The purpose of this study is to investigate how technology-driven approaches, such gamification and reward systems, might help university students waste less food and packaging. Whether social motivation and minor rewards can promote long-term behavior change is the main query. A cleaner campus, a more sustainable environment, and a student body that feels empowered to take ownership of their actions in support of a greener future are the results of a successful initiative.

Research Question

How can technology-driven awareness and reward systems reduce food and packaging waste behavior among university students?

Research Objective

The goal of this study is to look into the behaviors, mindset, and use of technology when it comes to food and packaging waste among university students. We also want to explore how digital tools might help solve this problem. Specifically, the main objectives are:

1. To understand the key behavioral reasons behind food and packaging waste on campus.
2. To see how technology and reward systems could encourage students to adopt more sustainable habits.
3. To test if tech-driven awareness campaigns and rewards can actually reduce food and packaging waste in the university community.

By addressing these, this research hopes to offer practical ideas on how we can encourage more sustainable behavior using innovative, tech-based solutions that fit into university life.

Literature Review

Managing waste effectively is one of the most important parts of making university campuses more sustainable. But doing that is not as simple as it sounds. Universities have to deal with several challenges, like getting students to change their everyday habits, using the right technology, and setting up proper waste management systems. The research discussed below looks at different ways this issue has been approached and shows that there isn't a single solution. Real change needs a mix of policies, awareness efforts, and creative use of technology to actually work.

Source 1: A Study of the Behavioral and Environmental Factors Influencing Food Waste in Higher Education

This study explores why students waste food in the first place, looking at how personal habits, surroundings, and social factors affect their behavior [1]. The researchers found that things like gender, age, peer influence, and even personal traits such as BMI all play a part in how much food students end up throwing away. Basically, both individual actions and the environment students are in make a big difference. The paper gives a better understanding of why food waste happens at the student level and suggests ways to design more focused awareness campaigns to reduce it [1].

Source 2: FoodWise: Food Waste Reduction and Behavior Change on Campus with Data Visualization and Gamification

The second study talks about the FoodWise project, which used data visualization and gamification to get students more involved in reducing food waste [2]. The project had a live dashboard that showed food waste data from university canteens and a mobile app where students could log their food-saving actions to earn small rewards. During a two-week trial at HKUST, more than 200 students took part, recording over 800 food-saving actions every day [2]. The results showed that when students can actually *see* the impact of their choices and get rewarded for it, they're more likely to change their behavior and make sustainability a part of their routine.

Source 3: Waste Minimization Initiatives in Campus Sustainability: The Experience of Universiti Teknologi Malaysia

This paper focuses on how Universiti Teknologi Malaysia (UTM) integrated sustainability into everyday campus life using a "Living Lab" approach [3]. The university began by analyzing the types of waste being generated and discovered that nearly half was food waste, while over 40 percent was recyclable material [3]. Based on these findings, UTM introduced the Green Office program, which encouraged staff to reduce paper use and hold more eco-friendly meetings. These efforts paid off, as paper consumption dropped from 30 percent in 2011 to 58 percent in 2013 [3]. This example shows how good planning, consistency, and proper follow-up can turn sustainability ideas into real and measurable progress.

Synthesis of Ideas

All three studies make one thing clear: reducing campus waste needs a layered and realistic approach. Zen et al. (2014) highlight that strong institutional support and clear sustainability policies form the foundation of any successful waste management plan [3]. Bittner et al. (2025) emphasize that understanding what influences student behavior is essential for making those plans actually work [1]. Yu et al. (2023) show that technology and gamification can help turn awareness into real action [2]. In the end, the most effective campus waste management systems are the ones that combine these three elements: institutional backing, behavioral insight, and interactive technology that encourages students to build lasting sustainable habits.

Survey Summary

The survey results give a pretty clear picture of how common food and packaging waste has become on campus, something most students have probably noticed themselves. A large number of respondents **(70.6%)** said they “often” or “always” see leftover food or packaging lying around in cafeterias or outdoor areas, showing that it is not just an occasional problem. On the other hand, **64.7%** claimed they “always” throw their waste away properly, which suggests that many students do try to act responsibly. Still, about **29.4%** admitted that they sometimes do not dispose of waste correctly, which means that awareness alone is not solving the issue.

Some of the main reasons students gave for this behavior were laziness, lack of motivation, and not having easy access to bins. Almost half **(47.1%)** said the bins are too far away, and **23.5%** said they are often full, making it inconvenient to throw things away. This shows that better placement and maintenance of bins could really help. There is also a social side to the problem since many students feel that if no one faces any consequences for littering and everyone else is doing it too, then it is not a big deal. That kind of thinking makes it harder to encourage change.

Interestingly, most students see technology as part of the solution. Over half **(52.9%)** believe that tech tools like smart bins, mobile apps, or digital campaigns could help reduce waste on campus. The same percentage also said that simply having bins placed in more visible spots would encourage them to dispose of their waste properly. This matches another finding where **54.9%** said visibility of bins would be the biggest motivator for correct disposal. A smaller group (**29.4%**) thought that things like digital reminders or awareness campaigns could also help.

When asked if they agree that technology can help reduce food and packaging waste, **35.3%** agreed and **29.4%** strongly agreed. In addition, **76.5%** said student accountability is very important in keeping the campus clean, showing that most students understand this problem requires a collective effort rather than relying only on university policies.

Question	Question Type	Most Common Response	% of Respondents	Key Observation
How often do you notice leftover food or packaging (e.g., wrappers, cups) left on tables in cafeterias or outdoor spaces?	Likert Scale	Always	41.2%	Leftover waste is frequently observed by students. There seems to be a prevalent issue with waste left in public spaces.
How often do you personally make sure to properly dispose of food and packaging waste after eating on campus?	Likert Scale	Always	64.7%	A strong majority of students ensure they dispose of waste properly, though a few do so occasionally or rarely.
What are the main reasons students don't dispose of waste properly? (Select up to 2)	Multiple choice	Laziness / Lack of motivation	25%	The main reasons for improper disposal include a lack of motivation, peer influence, and absence of consequences.
Which of the following would most motivate you to dispose of waste responsibly?	Multiple choice	More visible bins / better bin placement	52.9%	Improved infrastructure (e.g., visible bins) is the most motivating factor for students to dispose of waste properly.
How strongly do you agree with the statement: "Technology (e.g., smart bins, apps, campaigns) could help reduce food and packaging waste behavior on campus."	Likert Scale	Neutral	35.3%	Students have mixed opinions, with a good number being neutral, suggesting varying degrees of faith in technology's role.
How important do you think it is for students to take responsibility for keeping campus spaces clean?	Likert Scale	Extremely important	76.5%	There is a strong belief in the importance of student responsibility for maintaining campus cleanliness.

Table 1: Summary of survey findings

Interview Insights

Looking at both interviews, it's pretty clear that most students don't think food waste on campus can ever be fully solved, though they believe it can be reduced if people become more aware and responsible about what they consume. One student said, *"I don't think food wastage can ever be fully solved, but it can definitely be improved with more awareness and better control,"* while the other agreed, saying it can't be completely fixed but can be managed *"if both the university and students try to plan things better and be more aware."* Both also talked about portion control, suggesting that students should only take what they can actually eat instead of piling up food and wasting it. They felt that awareness makes a big difference because once people understand the impact of their actions, they're more likely to change. Both agreed that while the university can help through reminders or awareness campaigns, *"the main responsibility still lies with students since they're the ones creating the waste in the first place."*

They also pointed out that changing habits and mindsets takes time. One mentioned that *"if wasting food or littering became socially unacceptable, people would start behaving differently,"* which shows how peer pressure and social norms can be more effective than strict rules. Both agreed that awareness and subtle social pressure could work better than enforcement.

When asked about using technology or reward-based systems to help reduce waste, they had slightly different views. One suggested that *"maybe something like an app or a digital board that tracks how much waste is being reduced"* could motivate students if small rewards or recognition were involved. The other disagreed, saying *"apps or points wouldn't really work long term"* but that *"recognition would probably work much better… since who doesn't wanna be at the top of the leaderboard?"* Both agreed that tech could make awareness more interactive but admitted it would only succeed if the university supported and maintained it properly.

On the question of responsibility, one student believed it's *"mostly on the students"* since the café can't control people's behavior, while the other said it should be shared, with the university guiding and maintaining cleanliness but students taking initiative. Both noticed leftover food and wrappers in the café but appreciated that the cleaning staff handled it quickly.

Overall, both interviews showed that students care about the issue and see it as something that depends more on personal behavior than on university rules. They believe awareness, mindfulness, and a bit of social motivation can go a long way. And while food waste can't be completely eliminated, it can definitely be reduced with real effort from both students and the university. The complete interview questions and answers are included in **Appendix B** for reference.

Problem Brief

If you've ever walked through the cafe after class, you've probably seen plates of leftover chutney, empty bottles, and enough paper plates to start a recycling plant. Food and packaging waste has quietly become one of those "everyone knows, no one fixes" problems on campus. It's not just gross, it also shows how conveniently we ignore responsibility. We make the mess, pay for the cleanup, and still talk about sustainability like we actually care.

The worst affected are hostelites and students who basically live on campus. Cafeterias hand out disposable containers like candy, and no one has time or energy to clean up. Posters and "Go Green" drives fade faster than mid-semester motivation.

The real problem isn't the waste, it's the behavior. So what if sustainability was actually fun? Imagine an app that tracks your eco-actions and gives you bragging rights on a leaderboard. If turning off notifications feels rewarding, maybe reusing containers can too. The challenge is finding a way to make sustainability effortless and even rewarding. That's where technology and incentive-based systems might make a difference.

References

[1] B. Bittner, V. Vida, D. Szakos, G. Kasza, and S. Kovács, "A study of the behavioral and environmental factors influencing food waste in higher education," *Cleaner Waste Systems*, vol. 11, p. 100256, 2025. doi: 10.1016/j.clwas.2025.100256

[2] Y. Yu, S. Yi, X. Nan, L. Y.-H. Lo, K. Shigyo, L. Xie, J. Wicaksana, K.-T. Cheng, and H. Qu, "FoodWise: Food waste reduction and behavior change on campus with data visualization and gamification," in *Proc. ACM SIGCAS/SIGCHI Conf. Computing and Sustainable Societies (COMPASS '23)*, 2023, pp. 1–8. doi: 10.1145/3588001.3609364

[3] I. S. Zen, D. Subramaniam, H. Sulaiman, R. Ahamad, W. Omar, A. L. Saleh, and A. A. Rahman, "Waste minimization initiatives in campus sustainability: The experience of Universiti Teknologi Malaysia," in *Proc. 2nd Symp. Urban Mining (SUM 2014)*, 2014, pp. 1–10.

Appendix A - Survey Results

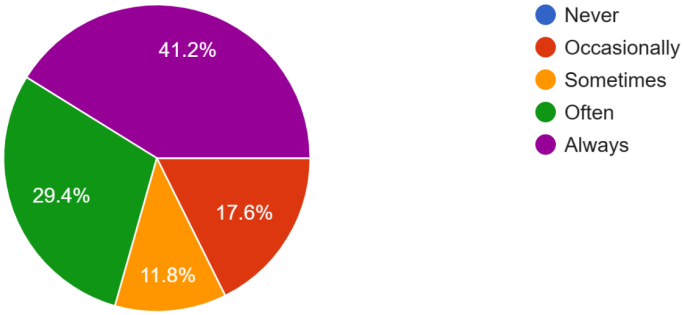
Timestamp	How ofte	How often	What are the main reasons	Which of the following would	How strongly do "Technology (e.g	How important do you th
10/7/2025 7:53:09	Always	Always	No consequences for leaving wa	Fines or strict rules	Neutral	Extremely important
10/7/2025 7:56:22	Sometimes	Always	Bins are too far / not visible, Bins	More visible bins / better bin pla	Strongly agree	Extremely important
10/7/2025 8:41:34	Often	Always	"Everyone else does it" mentality	Fines or strict rules	Neutral	Extremely important
10/7/2025 8:46:43	Always	Always	Laziness / lack of motivation, Bi	More visible bins / better bin pla	Strongly agree	Extremely important
10/7/2025 8:47:30	Occasionally	Always	Laziness / lack of motivation, "E	Reminders or digital campaigns	Agree	Extremely important
10/7/2025 8:55:39	Always	Often	Laziness / lack of motivation, No	More visible bins / better bin pla	Agree	Extremely important
10/7/2025 9:03:47	Often	Often	Laziness / lack of motivation, Bi	More visible bins / better bin pla	Agree	Moderately important
10/7/2025 9:04:25	Sometimes	Always	Laziness / lack of motivation, No	Fines or strict rules	Neutral	Extremely important
10/7/2025 16:59:29	Always	Always	Laziness / lack of motivation, Bi	More visible bins / better bin pla	Agree	Extremely important
10/7/2025 19:09:17	Always	Always	Laziness / lack of motivation, "E	Fines or strict rules	Neutral	Extremely important
10/7/2025 19:50:50	Often	Often	Laziness / lack of motivation, "E	More visible bins / better bin pla	Strongly agree	Extremely important
10/8/2025 14:45:08	Always	Often	Bins are often full	Reminders or digital campaigns	Disagree	Moderately important
10/8/2025 14:45:26	Occasionally	Sometimes	"Everyone else does it" mentality	Fines or strict rules	Neutral	Slightly important
10/8/2025 15:33:50	Occasionally	Always	Laziness / lack of motivation, No	More visible bins / better bin pla	Strongly agree	Extremely important

Responses from Google Sheet

Data Visualization:

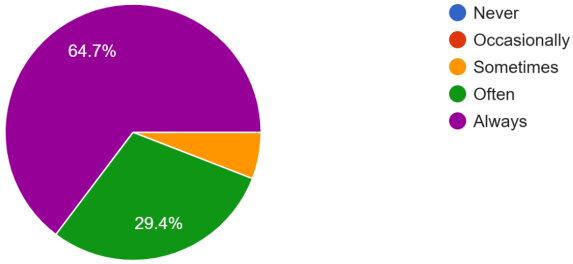
How often do you notice leftover food or packaging (e.g., wrappers, cups) left on tables in cafeterias or outdoor spaces?

17 responses



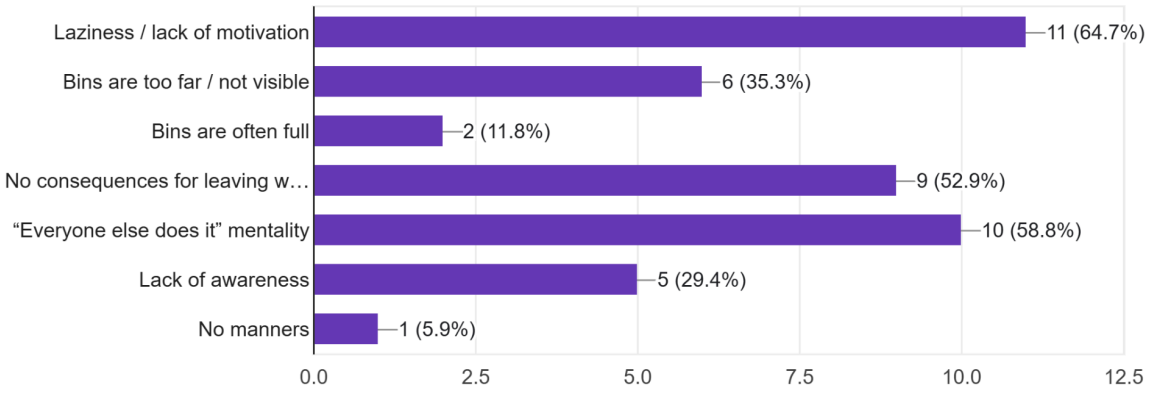
How often do you personally make sure to properly dispose of food and packaging waste after eating on campus?

17 responses



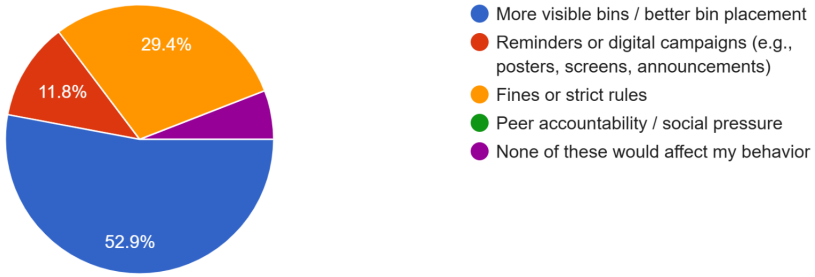
What are the main reasons students don't dispose of waste properly? (Select up to 2)

17 responses



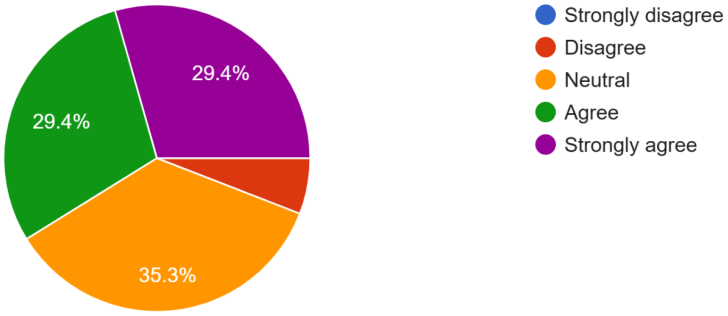
Which of the following would most motivate you to dispose of waste responsibly? (Select one)

17 responses



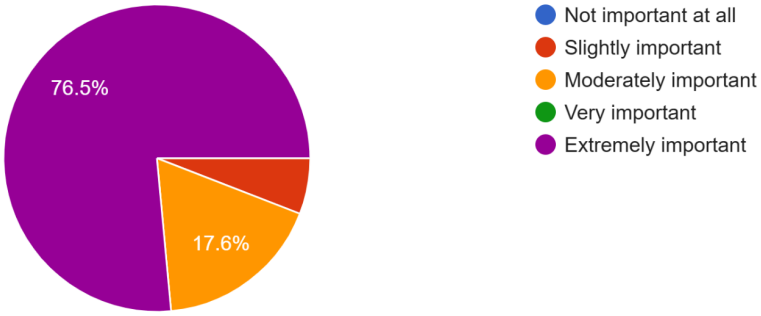
How strongly do you agree with the statement: "Technology (e.g., smart bins, apps, campaigns) could help reduce food and packaging waste behavior on campus."

17 responses



How important do you think it is for students to take responsibility for keeping campus spaces clean?

17 responses



Appendix B – Interview Notes

Interview 1:

Date: 6th October, 2025
Mode: In-person
Interviewer: Kabeer Ali
Interviewee: Fast Student

Q1. Do you think food wastage in university is an issue that can be fully resolved?
A1: Honestly, I don't think food wastage can ever be fully solved, but it can definitely be improved with more awareness and better control.

Q2. What steps can an individual take to diminish food wastage?
A2: I think people should just take only what they need and try not to waste food. Small things like that can actually help a lot.

Q3. Can we create a society where it's socially unacceptable to waste food or litter? If yes, then how?
A3: Yeah, I think we can, but only if people start taking it seriously. If we raise awareness and build a stronger campus culture around it, it could become something people actually care about.

Q4. How much of the responsibility lies on the organization (university) and how much on the students?
A4: It should go both ways. Students need to be more mindful, but the university can also help by organizing awareness campaigns or putting up reminders.

Q5. How often do you find wrappers or leftover food in the café?
A5: Pretty often. You can usually spot food and wrappers lying around, especially right after classes start as people just leave their stuff on the tables in a hurry.

Q6. Do you think using technology or a reward system (like apps, points, or recognition) could help reduce food and packaging waste among students? If yes, how?
A6: I think it could help a bit. Maybe something like an app or a digital board that tracks how much waste is being reduced or how many people are participating. If students got small rewards or even just recognition for being responsible, it might motivate them to waste less. But it would only work if people actually cared about it and the university was willing to sponsor all the rewards you'd be promising people through that app.

Interview 2

Date: 6th October, 2025
Mode: In-person
Interviewer: Muneeb Lone
Interviewee: Fast Student

Q1. Do you think food wastage in university is an issue that can be fully resolved?
A1: Not really. It's not something that can be completely fixed because everyone has different eating habits. But it can be reduced if both the university and students try to plan things better and be more aware.

Q2. What steps can an individual take to diminish food wastage?
A2: It mostly comes down to being mindful. Students should take only what they can actually eat and avoid wasting food just because they can.

Q3. Can we create a society where it's socially unacceptable to waste food or litter? If yes, then how?
A3: Maybe, but not right away. Changing people's mindset about these things takes time and effort. You can't expect it to happen overnight.

Q4. How much of the responsibility lies on the organization (university) and how much on the students?
A4: Honestly, it's mostly on the students. The university can't really control how people behave in the café. It's up to each person to handle their own waste properly.

Q6. Do you think using technology or a reward system (like apps, points, or recognition) could help reduce food and packaging waste among students? If yes, how?
A6: I don't really think that apps or points would work and even if they did I don't think they'd last in the long term, plus they'd be extremely hard to implement because as students, how are you ever going to fund a rewards system? You would need the university to pitch in at that point and after being here for 3 years, I seriously doubt that they would. However, recognition would probably work much better in my personal opinion since who doesn't wanna be at the top of the leaderboard? That kind of thing would give it a twist of healthy competition.

Notes on Discussion about Food Waste on Campus

Both students agreed that food wastage on campus can't be entirely eliminated, but it can definitely be reduced through a combination of **awareness, mindfulness, and personal responsibility**.

- *Student Opinions on Responsibility:*

    - **P1** believed that responsibility should be shared between both students and the university. He/she suggested that the university could play a pivotal role by running ***awareness campaigns*** or providing ***reminders*** to students about proper waste disposal.
    - **P2**, on the other hand, thought that the issue largely depends on students' own behavior and habits, emphasizing that personal responsibility is crucial.

- *Views on Technology and Reward Systems:*
  When asked about the use of ***technology or reward systems*** to reduce food waste, the students' views diverged:

    - **P1** was in favor of ***apps or digital boards*** that show ***waste reduction*** **progress** and offer small rewards or recognition. However, he/she emphasized that these solutions would only be effective if students genuinely care about the issue in the long term. Additionally, he/she believed that the university would need to support such rewards for the system to work.
    - **P2** disagreed, stating that point-based systems would likely not work in the long run. Instead, he/she argued that ***public recognition*** and ***leaderboards*** could help create ***healthy social pressure,*** motivating students to act more responsibly.

- **General Observations**:

    - Both students acknowledged that leftover food and wrappers were frequently observed in the campus **café**. However, they also appreciated the cleaning staff's **quick response** in managing the waste, demonstrating that while maintenance is efficient, there is still a need for improvement in **student behavior**.

In conclusion, while students agree that waste reduction is possible, it requires a collective effort from both individuals and the university. Introducing technology or reward systems, while debated, could serve as an additional motivation, but the primary challenge remains shifting student behavior toward sustainability.